MAIS PAC

IPhO 02.11: Optics: Lenses, Gratings and Interferometers · Worked Solutions

Optics: Lenses, Gratings and Interferometers: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam’s constants take precedence.

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Solution 1 — Refraction into glass. Return →

(a) 1.00sin 40° = 1.50sin θ_2, so sin θ_2 = 0.643/1.50 = 0.429 and θ_2 = 25.4° — the ray bends towards the normal.

(b) v = c/n = 3.00×10^8/1.50 = 2.00×10^8 {m,s^(-1)}, and λ = λ_0/n = 600/1.50 = 400 {nm}.

(c) sin θ_c = 1/1.50 = 0.667, so θ_c = 41.8°. **Check:** the frequency is c/λ_0 = 5.0×10^(14) {Hz} in both media, and v/λ gives the same value, as it must.

Solution 2 — A converging lens. Return →

(a) (1)/(v) = (1)/(15) - (1)/(40) = (8-3)/(120) = (5)/(120), so v = 24 {cm}.

(b) m = -v/u = -24/40 = -0.60, so h_i = -0.60×3.0 = -1.8 {cm}: the image is inverted and 1.8 {cm} tall.

(c) Real, inverted, diminished — the camera and the human eye use exactly this case, with the object beyond 2f. **Check:** v = 24 {cm} lies between f = 15 and 2f = 30, consistent with the table for u>2f.

Solution 3 — The magnifying glass. Return →

(a) (1)/(v) = (1)/(8) - (1)/(6) = (3-4)/(24) = -(1)/(24), so v = -24 {cm} and m = -v/u = +4.0.

(b) v is negative, so the image is **virtual**, on the same side as the object, upright and four times larger. You look **through** the lens, with your eye on the far side, and the image appears to be 24 {cm} behind the lens.

(c) M = 25/f = 25/8.0 = 3.1 for a relaxed eye. **Check:** the virtual image is further away than the object, which is why the angular size grows even though the lens does not add any light.

Solution 4 — Total internal reflection and optical fibres. Return →

(a) sin θ_c = (n_({clad}))/(n_({core})) = (1.48)/(1.52) = 0.974, so θ_c = 77.0° measured from the normal to the core–cladding boundary, i.e. only 13° from the fibre axis.

(b) A ray entering at angle α to the axis is refracted to angle θ with sin α = n_({core})sin θ, and must then hit the wall at 90°-θ > θ_c. The limiting case gives sin α_(max) = √(n_({core})²-n_({clad})²) = √(2.310-2.190) = 0.346, so α_(max) = 20° — a numerical aperture of 0.35.

(c) A thin fibre keeps the ray meeting the wall at many small angles of incidence rather than letting the beam wander; more importantly, a thin core makes the fibre single-mode, so all rays travel nearly the same path length and a pulse does not smear out. **Check:** the index difference is small (2.6%), which is what makes θ_c so close to grazing — a large index step would guide more light but spread the pulse badly.

Solution 5 — Young's double slit. Return →

(a) y = (λ L)/(d) = ((589×10^(-9))(2.0))/(0.40×10^(-3)) = 2.9×10^(-3) {m} = 2.9 {mm}.

(b) y_3 = 3y = 8.8 {mm}.

(c) By definition y_3 is where Δ = 3λ; explicitly, Δ = d,y_3/L = (0.40×10^(-3))(8.84×10^(-3))/2.0 = 1.77×10^(-6) {m} = 3×589 {nm}. **Check:** the fringe spacing is small enough that the small-angle approximation is excellent, with θ_3 = 8.8×10^(-3)/2.0 = 4.4 {mrad}.

Solution 6 — A diffraction grating. Return →

(a) d = 1/(500 {mm^(-1)}) = 2.0×10^(-6) {m}.

(b) sin θ_m = mλ/d = m(600×10^(-9))/(2.0×10^(-6)) = 0.300m: θ_1 = 17.5° and θ_2 = 36.9°.

(c) For m=3, sin θ = 0.90, which is **less** than 1 — so a third order does exist at θ = 64.2°. The highest order is given by sin θ1: m d/λ = 3.33, so orders m = 0, 1, 2, 3 appear, and at m=3 the beam leaves at 64°, close to the plane of the grating, where it is very faint. **Check:** the largest order is always d/λ, and here the integer part of 3.33 is 3.

Solution 7 — Two lenses in contact. Return →

(a) (1)/(f) = (1)/(20) + (1)/(-30) = (3-2)/(60) = (1)/(60), so f = +60 {cm} and P = 1/0.60 = +1.67 {D} (the powers are +5.0 {D} and -3.3 {D}).

(b) (1)/(v) = (1)/(60) - (1)/(50) = (5-6)/(300), so v = -300 {cm}: a virtual image 3.0 {m} on the object's side, with m = -v/u = +6.0.

(c) Powers add because they are the reciprocals of focal lengths, and thin lenses in contact share the same incoming and outgoing rays: their deflections add. Focal lengths would combine through the awkward harmonic rule (1)/(f) = (1)/(f_1)+(1)/(f_2). **Check:** the combination is weaker than either lens, since 60 {cm} > 20 {cm} — a diverging lens always weakens a converging one.

Solution 8 — A thin film. Return →

(a) At the air–film surface the reflection has a π phase change; at the film–air surface it does not. Constructive interference therefore needs

$$2nt = (m+(1)/(2))\lambda \quad t = (\lambda)/(4n) = (550\ \{nm\})/(4(1.33)) = 103\ \{nm\} \qquad (1)$$

for the thinnest film.

(b) Solve 2nt = (m+(1)/(2))λ for λ with 2nt = 2(1.33)(103 {nm}) = 274 {nm}: λ = 274/(m+0.5) gives 548 {nm} for m=0, 183 {nm} for m=1 — so no second visible wavelength; the two wavelengths that would match are ultraviolet. (A thicker film would show several visible colours at once.)

(c) As the film thins to a few nanometres, 2nt≪λ for all visible light and the condition becomes destructive for every wavelength, so nothing is reflected: a black film. **Check:** this is exactly the design of an anti-reflection coating, with t = λ/(4n).

Solution 9 — The Rayleigh criterion. Return →

(a) θ_(min) = 1.22(λ)/(D) = 1.22(550×10^(-9))/(0.20) = 3.4×10^(-6) {rad}.

(b) s = θ_(min)×3.8×10^8 = 1.3×10³ {m} — about 1.3 {km}, so the telescope can resolve large features but not individual buildings.

(c) Magnification spreads the same blurred image over a larger angle; it adds no information. The resolution is set by diffraction at the aperture, so only a larger D (or a shorter λ) reduces θ_(min) — which is why astronomy pushes to bigger mirrors and why radio telescopes must be enormous. **Check:** doubling the aperture halves θ_(min), and a 0.20 {m} telescope is close to the practical limit set by atmospheric seeing.

Solution 10 — A prism at minimum deviation. Return →

(a) With r_1 + r_2 = A and D = i_1 + i_2 - A, differentiating D with respect to i_1 and setting dD/di_1 = 0 gives di_1 = di_2, which together with Snell's law forces i_1 = i_2 and hence r_1 = r_2 = A/2.

(b) r_1 = 30°, so sin i_1 = 1.50sin 30° = 0.750 and i_1 = 48.6°. Then

$$D_{(\min)} = 2i_1 - A = 97.2^\circ - 60^\circ = 37.2^\circ. \qquad (2)$$

(c) The refractive index depends on wavelength, so each colour has its own D_(min) — that is dispersion, and the prism spreads white light into a spectrum. A grating does it better because its dispersion dθ/dλ = m/(dcos θ) is much larger and is nearly uniform in wavelength, whereas a prism's is nonlinear and limited by the material. **Check:** at minimum deviation the ray inside the prism is parallel to the base — the standard laboratory way of setting a prism on a spectrometer.